Time Series Exercise Sheet 11

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Exercise 11.1

The AIC and corrected AIC are given by

$$\begin{aligned} \mathrm{AIC}(\boldsymbol{\theta}) &= -2\ln L(\boldsymbol{\theta}|\boldsymbol{y}) + 2k, \\ \mathrm{AICC}(\boldsymbol{\theta}) &= -2\ln L(\boldsymbol{\theta}|\boldsymbol{y}) + 2k\frac{n}{n-k-1}, \end{aligned}$$

where n is the sample size and k is the number of estimated parameters.

Explain under what conditions on n and k the two criteria become approximately equivalent.

Solution 11.1

Subtracting AIC from AICC gives

$$\begin{aligned} \mathrm{AICC}(\boldsymbol{\theta}) - \mathrm{AIC}(\boldsymbol{\theta}) &= 2k\left(\frac{n}{n-k-1} - 1\right) \\ &= 2k\left(\frac{n-(n-k-1)}{n-k-1}\right) \\ &= 2k\left(\frac{k+1}{n-k-1}\right) \\ &= \frac{2k(k+1)}{n-k-1}. \end{aligned}$$

Thus the two criteria are approximately equivalent when this correction term is small. In particular, if k is fixed and $n \to \infty$, then

$$\frac{2k(k+1)}{n-k-1} \to 0,$$

so AICC and AIC become equivalent. More generally, if $k = k_n$ is allowed to depend on n, the correction is small when n is large relative to k_n^2, for example when $k_n^2/n \to 0$ as $n \to \infty$. If k grows too quickly with n, the correction need not vanish, and AICC can remain meaningfully different from AIC.

Exercise 11.2

Determine the form of AICC for a mean-zero ARMA(p, q) model when exactly m of the AR and MA coefficients are known in advance and fixed to zero.

Solution 11.2

If exactly m of the $p + q$ AR and MA coefficients are fixed to zero, then only $p + q - m$ of those coefficients are estimated. Including the innovation variance, the total number of estimated parameters is therefore

$$k = p + q + 1 - m.$$

Substituting this value of k into the AICC formula gives

$$\mathrm{AICC}(\boldsymbol{\theta}) = -2\ln L(\boldsymbol{\theta}|\boldsymbol{y}) + 2(p+q+1-m)\frac{N}{N-(p+q+1-m)-1}. \tag{1}$$

Exercise 11.3

For a time series of $n = 150$ observations, a mean-zero ARMA$(1,1)$ model is fitted and the sample residual autocorrelations at lags 1 to 4 are

$$\hat{r}_1 = 0.03, \quad \hat{r}_2 = -0.09, \quad \hat{r}_3 = 0.11, \quad \hat{r}_4 = -0.05.$$

(a) Compute the Ljung-Box statistic Q_4.

(b) The null hypothesis $H_0 : \rho_1 = \rho_2 = \cdots = \rho_4 = 0$ is tested at the 5% significance level. Under H_0, the statistic follows a χ^2_{m-p-q} distribution. State the degrees of freedom, use R to find the critical value, and give your conclusion.

(c) Would your conclusion change if instead you had fitted an AR(1) model? Explain briefly.

Solution 11.3

(a) The Ljung-Box statistic is

$$Q_m = n(n+2)\sum_{\tau=1}^{m}\frac{\hat{r}_\tau^2}{n-\tau}.$$

With $n = 150$ and $m = 4$:

$$\begin{aligned}
Q_4 &= 150 \cdot 152 \left(\frac{0.03^2}{149} + \frac{(-0.09)^2}{148} + \frac{0.11^2}{147} + \frac{(-0.05)^2}{146}\right) \\
&= 22800 \left(\frac{0.0009}{149} + \frac{0.0081}{148} + \frac{0.0121}{147} + \frac{0.0025}{146}\right) \\
&= 22800 \left(6.04 \times 10^{-6} + 5.47 \times 10^{-5} + 8.23 \times 10^{-5} + 1.71 \times 10^{-5}\right) \\
&\approx 22800 \times 1.60 \times 10^{-4} \approx 3.65.
\end{aligned}$$

(b) For an ARMA$(1,1)$ with $p = 1$ and $q = 1$, the degrees of freedom are $m - p - q = 4 - 1 - 1 = 2$. The critical value at the 5% level is obtained in R via `qchisq(0.95, df = 2)` which gives 5.991. Since $Q_4 \approx 3.65 < 5.991$, we do not reject H_0. There is no significant evidence of residual autocorrelation; the ARMA$(1,1)$ model appears adequate.

(c) For an AR(1) model, $p = 1$ and $q = 0$, so the degrees of freedom become $4 - 1 - 0 = 3$. The critical value is `qchisq(0.95, df = 3)` which gives 7.815. Since $Q_4 \approx 3.65 < 7.815$, the conclusion would still be to not reject H_0. In general, fitting fewer parameters leaves more degrees of freedom and widens the acceptance region.

Exercise 11.4

The AIC and BIC for a model with k parameters fitted to n observations are

$$\mathrm{AIC} = -2\log L + 2k, \qquad \mathrm{BIC} = -2\log L + k\log n.$$

(a) Show that the BIC penalty exceeds the AIC penalty if and only if $n > e^2 \approx 7.4$.

(b) For a fixed log-likelihood, a model with $k = 3$ parameters competes against one with $k = 2$ parameters. For $n = 50$, which criterion is more likely to select the larger model? Justify your answer by comparing the penalty differences.

Solution 11.4

(a) The AIC penalty per parameter is 2 and the BIC penalty per parameter is $\log n$. We need

$$\log n > 2 \iff n > e^2 \approx 7.39.$$

For any realistic sample size ($n \geq 8$) the BIC imposes a larger penalty per parameter than AIC, meaning BIC favours more parsimonious models.

(b) With $k = 3$ versus $k = 2$ (one extra parameter), the extra penalty for AIC is $2 \times 1 = 2$, while for BIC it is $\log(50) \approx 3.91$. BIC requires the extra parameter to improve the log-likelihood by at least 1.96 log-units, whereas AIC only requires an improvement of 1. Thus AIC is more willing to select the larger model, in contrast, BIC is harder to satisfy.